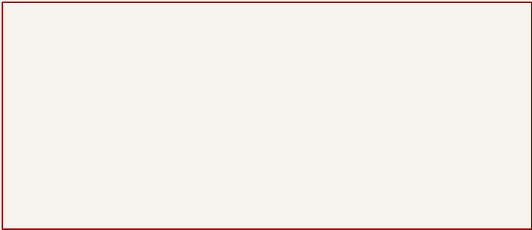


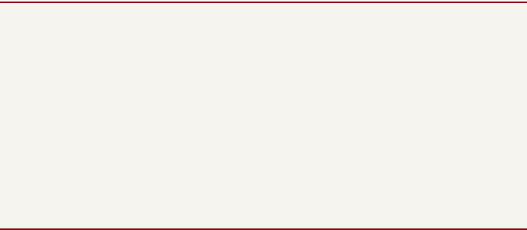
Vanchor Helm Board – sheet index

- 1 carrier: 12V in, protection, 5V buck header, Orange Pi Zero 3 ribbon header, UART/I2C JSTs
- 2 control: Pico 2, external thrust–driver interface (J13), BTN8982 servo bridge, AS5600

carrier



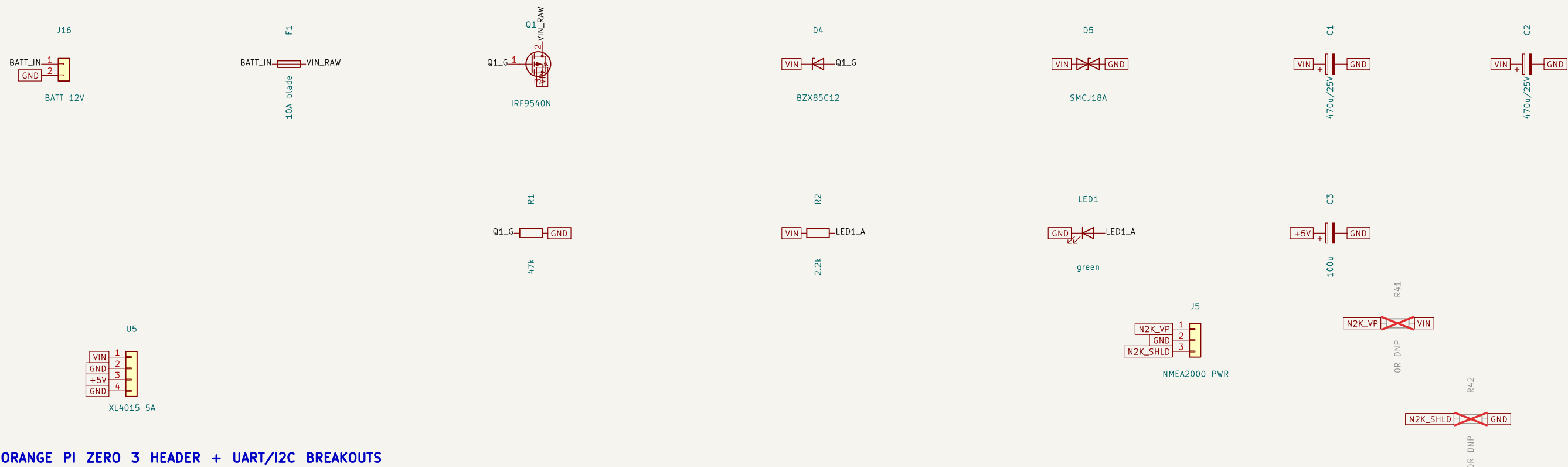
control



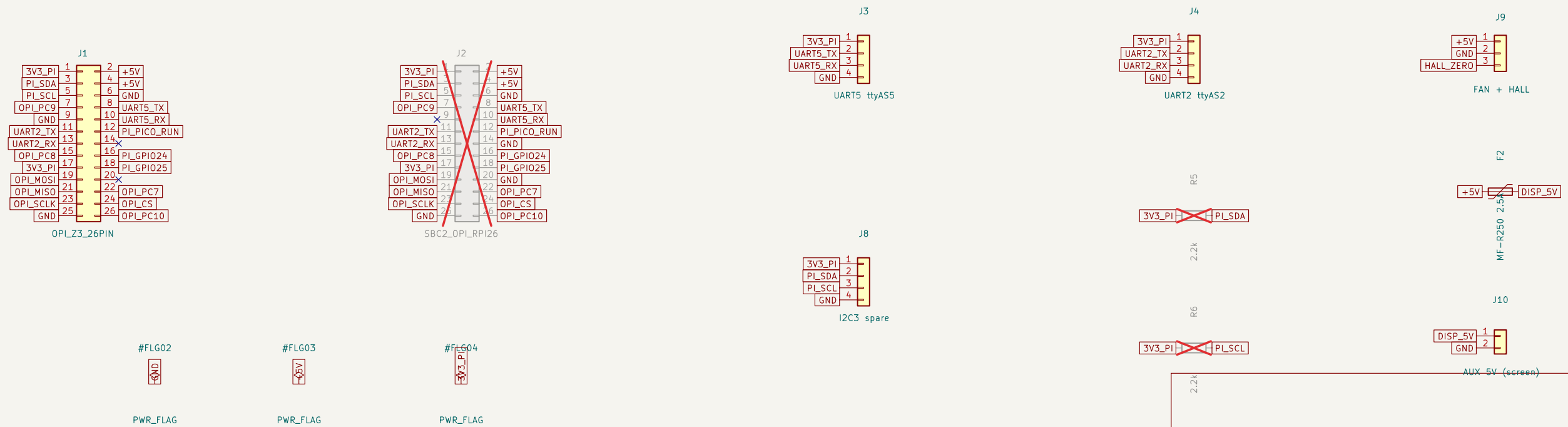
12V battery only; 5V from generic buck module on J14 Orange Pi Zero 3 (IDC ribbon) + Pico 2; external IBT–2 thrust driver; on–board servo bridge			
vanchor–ng			
Sheet: / File: vanchor–helm.kicad_sch			
Title: Vanchor Helm Board			
Size: A3	Date: 2026–07–02		Rev: v4.1
KiCad E.D.A. 10.0.3	Id: 1/3		

POWER: 12V battery -> F1 -> reverse-FET -> VIN; XL4015 buck daughterboard U5 (set 5.1V first)

CARRIER: J1 -> 26-way IDC ribbon -> Orange Pi Zero 3 (powered through the ribbon, no USB-C)



ORANGE PI ZERO 3 HEADER + UART/I2C BREAKOUTS



SERVO: 2x BTN8982TA on protected VIN; AS5600 encoder cable on J11

Sheet: /control/
File: control.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 3/3